

Musculoskeletal Trauma

Guideline Title: Musculoskeletal Trauma

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- I. **PURPOSE:** To recognize orthopedic injury including the ability to differentiate between closed and open fractures. After proper identification of the musculoskeletal injury, the provider must be able to treat and attempt to restore normal function of an injured extremity with a minimum of pain, deformity, and morbidity, in the least amount of time.

II. **PATHOPHYSIOLOGY:**

Injuries to the musculoskeletal system are a common occurrence in multisystem trauma. The primary concern is life-threatening hemorrhage or limb threatening injuries due to vascular compromise/traumatic amputation. Damage to large blood vessels, particularly arteries, can lead to major blood loss relatively quickly. Hemorrhage from long-bone fractures can also be significant.

Unexplained hypotension may be the only initial indication of major pelvic disruption with instability in the posterior ligamentous complex. The most important physical signs are progressive flank, scrotal, or perianal swelling and bruising, blood at the urethral meatus, and demonstrable mechanical instability. Manual manipulation of the pelvis should be performed only once during the physical examination. Repeated testing for pelvic instability may result in further hemorrhage. An indication of mechanical instability may be leg length discrepancy or rotational deformity (usually external) without a fracture of that extremity.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

- a. Assess and manage airway, breathing, and circulation.
- b. Obtain IV/IO access.
- c. If patient displays signs of inadequate perfusion, begin resuscitation with blood products per *GUID-PR025 - Emergency Blood Product Administration* and *GUID-PR026 - Warming of Blood Products & Intravenous Fluids*. For further resuscitation follow *GUID-M011 - Hypotension*.
- d. Control hemorrhage with well-aimed DIRECT PRESSURE and/or elevation of the extremity. If unable to control bleeding, consider:
 - i. Hemostatic Gauze application per *GUID-PR028 - Hemostatic Dressing - QuickClot Gauze*.
 - ii. Apply tourniquet per *GUID-PR006 Tourniquet Therapy*.
- e. Apply appropriate splint to avoid further injury, decrease discomfort and blood loss.

- f. Monitor vital signs closely.
- g. Monitor distal pulses, motor and sensation to the site of the injury. If pulses are compromised the extremity may be repositioned in an attempt to regain circulation.
- h. Treat pain and/or anxiety per *GUID-G005 - Analgesia and Sedation Management*.
- i. If amputation of extremity is involved:
 - i. Wrap the amputated part in saline soaked gauze and place in a watertight bag. Place the bag in ice filled container and transport with patient for possible reattachment. Do not soak amputated part directly in water.
 - ii. Wrap stump in sterile saline moistened dressing and apply pressure bandage.
- j. Provide pelvic immobilization if multisystem trauma from major MVC, motorcycle accident, ATV, bicycle accident, fall from >30 feet, or blast injury per *GUID-PR027 - Pelvic Circumferential Compression Binder*.
- k. Traction splints are indicated for fractures of the shaft of femur. Splints may be applied by the sending EMS crew and continued during transport.
 - i. Four Velcro support straps should be positioned 2 at mid-thigh, 1 above the knee, and 1 above the ankle. Never over the injury.
 - ii. The patient should feel relief from the pain.
 - iii. The injured foot should align with the uninjured foot.
 - iv. Reassess and record distal pulse, sensation, and motor function distal to the injury site.
- l. Consider TXA administration per *GUID-3203-T016 - Tranexamic Acid for Severe Hemorrhage*
- m. OPEN FRACTURES: If patient is stable and time permits, administer Cefazolin (Ancef) for antibiotic coverage. Mix each 1g via with 10ml Sterile water. Prioritize resuscitative efforts and pain control.
 - i. **Cefazolin (Ancef) Adult dose:**
 - 1. < 120 kg – 2 Grams Slow IV push over 3-5 minutes
 - 2. > 120 kg – 3 Grams Slow IV push over 3-5 minutes
 - ii. **Cefazolin (Ancef) Pediatric dose:**
 - 1. 25 mg/kg Slow IV push over 3-5 minutes (Maximum 2 Grams)
 - iii. Rapid push has been known to induce emesis and severe nausea
 - iv. Contraindications: cephalosporin allergy or penicillin allergy with a reaction that includes anaphylaxis, bronchospasm, or angioedema
 - v. Discontinue if signs of allergic reaction/anaphylaxis and treat per *GUID-M004 - Allergic Reaction Anaphylaxis* or *GUID-PED002 - Pediatric Allergic Reaction Anaphylaxis*
 - vi. Notify accepting facility of antibiotic administration, including time of administration

V. **DOCUMENTATION:** EMS charts

VI. **REFERENCES:**

- a. Bulger EM, Snyder D, Schoelles K, et al. (2014) An evidence-based prehospital guideline for external hemorrhage control: American College of Surgeons Committee on Trauma. *Prehosp Emerg Care*, 18: 163.